

Cloud-Based ERP Attendance Management System

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1. Abstract

The Cloud-Based ERP Attendance Management System is a comprehensive, multi-tenant web application engineered to modernize institutional record-keeping. Developed utilizing a high-performance Python backend via FastAPI and a reactive frontend using Alpine.js and Tailwind CSS, this system successfully replaces error-prone manual tracking processes. The architecture guarantees strict data isolation by dynamically provisioning database schemas for individual faculty members on a PostgreSQL cloud cluster (Supabase). Key features include algorithmic Excel data parsing for bulk operations, dynamic PDF report generation utilizing ReportLab, and an integrated student leave application pipeline. This project significantly enhances data integrity, real-time analytics, and administrative workflow efficiency.

2. Technology Stack & Dependencies

The system is built upon a modern, robust technology stack ensuring high scalability and seamless user experience:

Backend Framework: Python 3.x, FastAPI (for asynchronous API routing), and Uvicorn (ASGI server).

Frontend Interface: HTML5, Tailwind CSS (utility-first styling), and Alpine.js (lightweight state management).

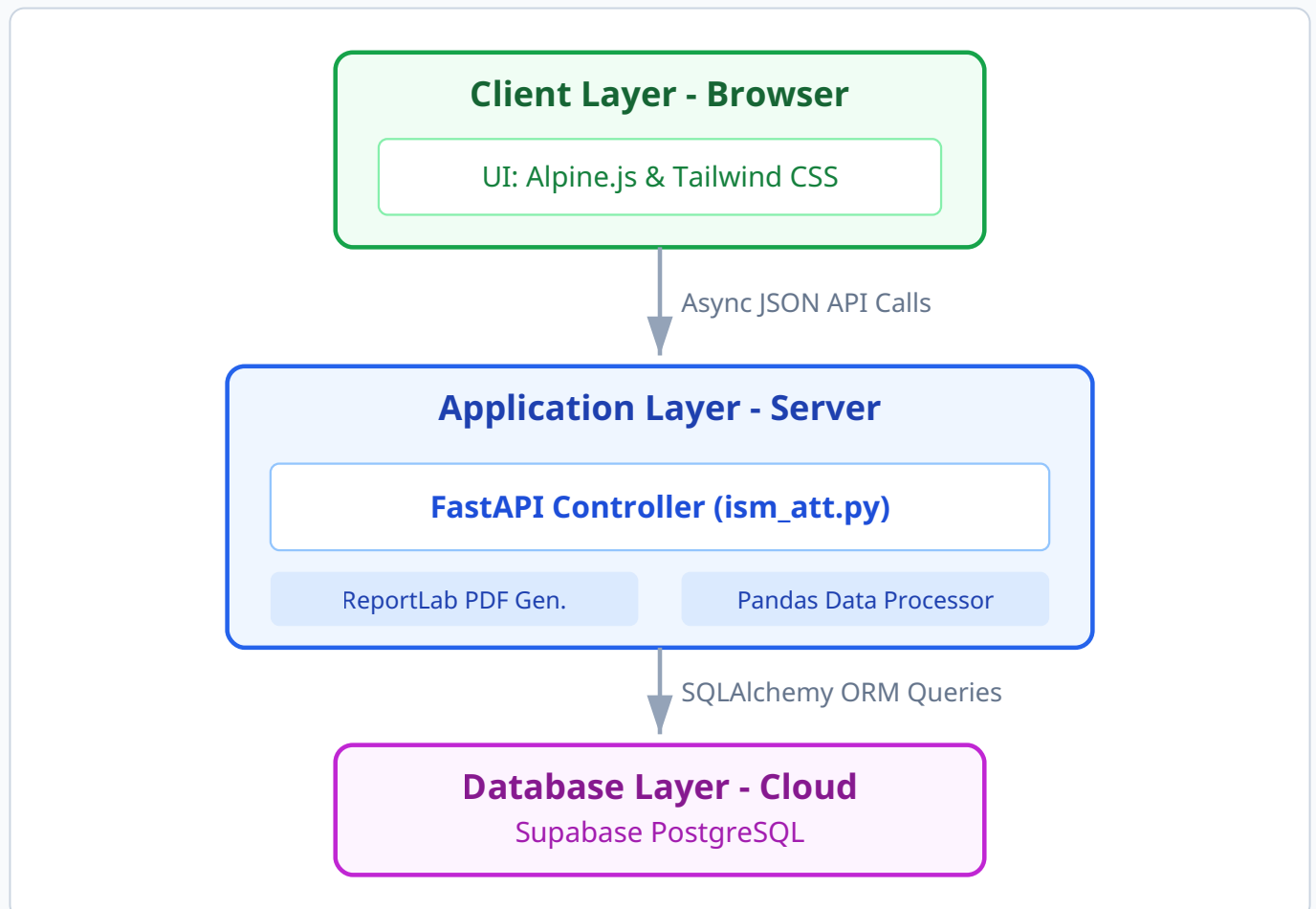
Database Management: PostgreSQL hosted on Supabase, integrated via SQLAlchemy (Object-Relational Mapping) and psycopg2.

Data Processing Modules: Pandas and Openpyxl for handling and parsing .xlsx and .csv files.

Document Generation: ReportLab framework for constructing automated, tabular PDF documents.

3. System Architecture Diagram (MVC Model)

The application implements a strict Model-View-Controller (MVC) and Client-Server architecture.



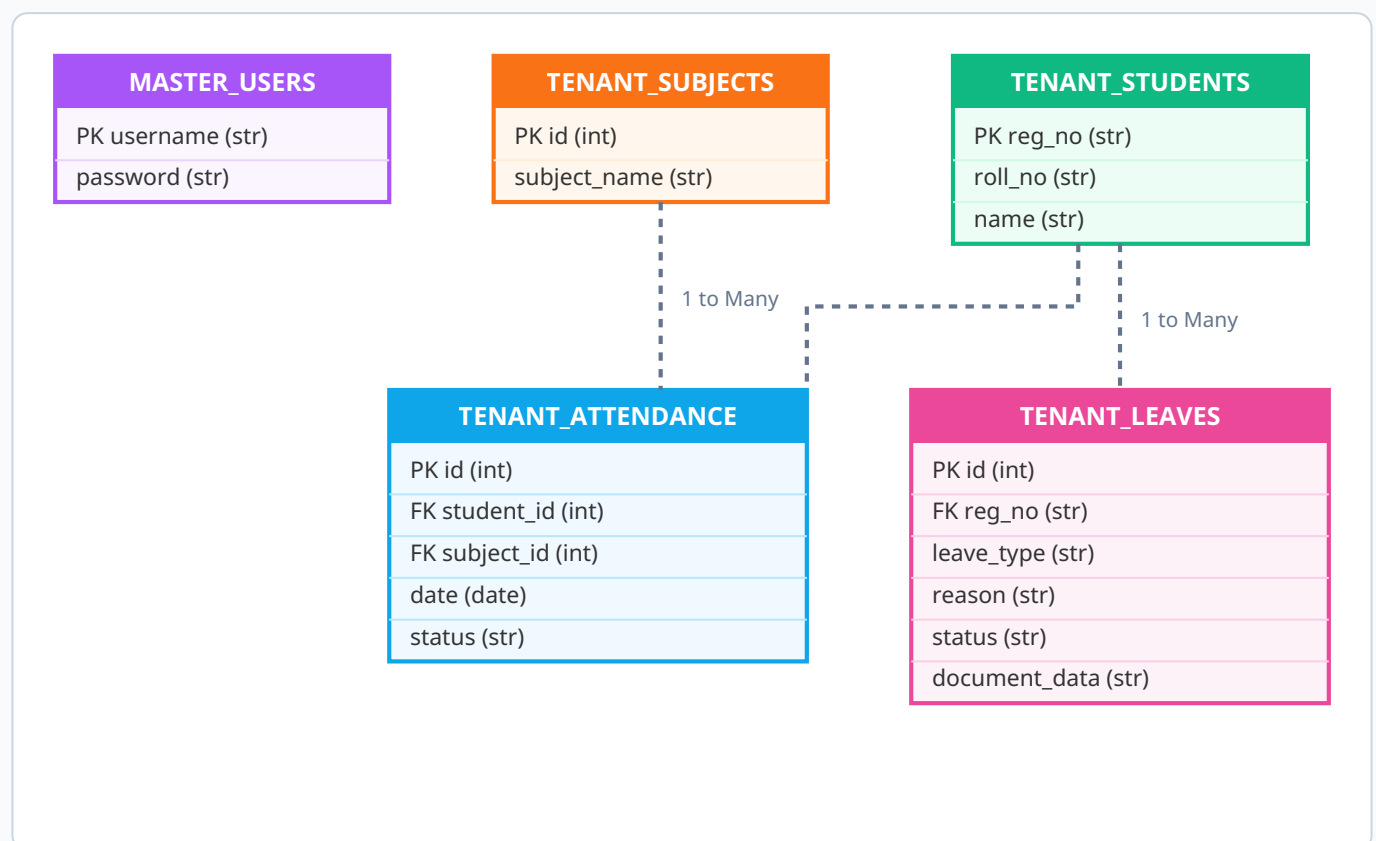
Client Layer: Captures faculty/student inputs asynchronously without page reloads.

Application Layer: Validates authentication, processes business logic, and compiles binary files (PDFs/Excel).

Data Layer: Manages persistent storage and structured relational mapping.

4. Entity-Relationship (ER) Diagram Schema

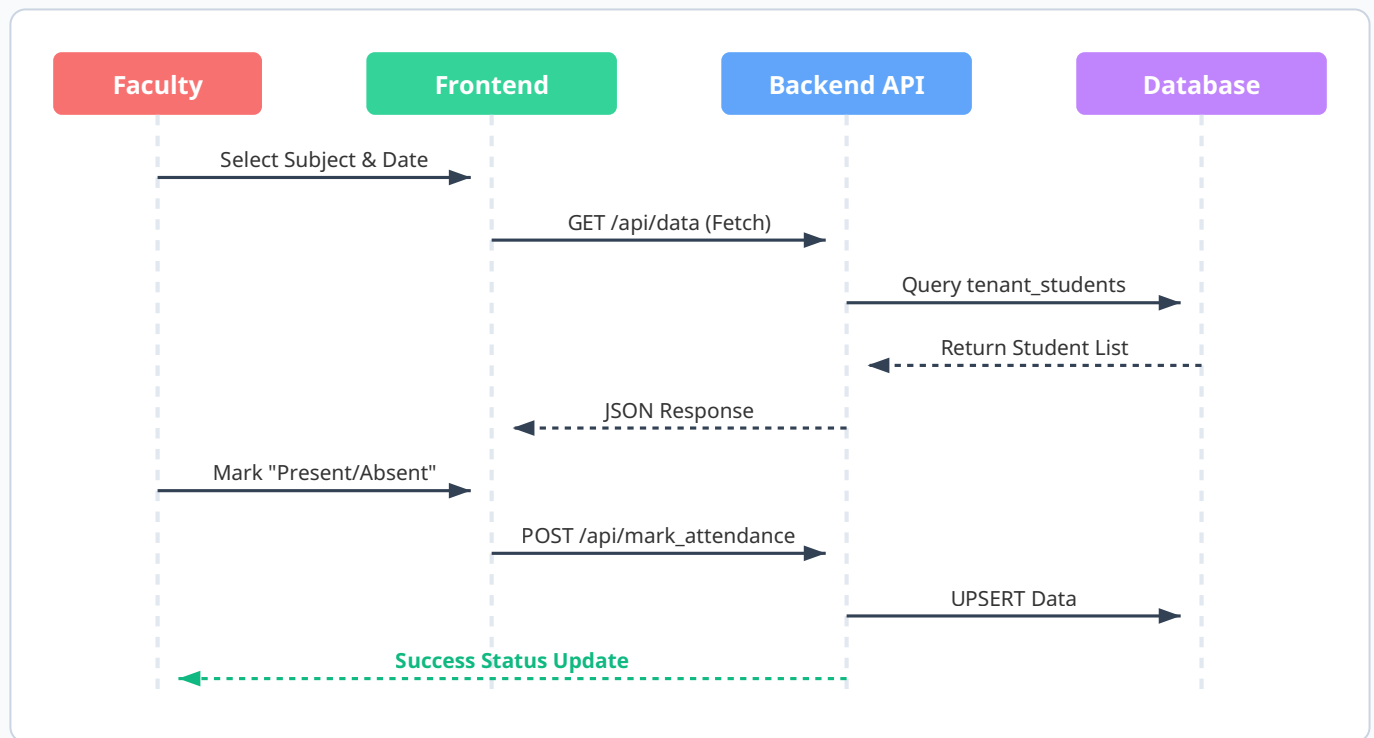
To support isolated, multi-tenant functionality, the system generates unique tables prefixed with the user's ID ({tenant}_tablename).



Relational Integrity: The attendance table strictly maps the student_id and subject_id using Foreign Keys, and enforces a unique constraint on (student_id, subject_id, date) to prevent duplicate records.

5. Data Flow Diagram (DFD) - Core Processes

DFD Level 1: Marking Daily Attendance Workflow



6. Implementation Highlights

Automated PDF Reporting: The backend leverages ReportLab to dynamically construct scalable, multi-column tables. It algorithmically calculates total classes and individual attendance percentages, embedding them into stylized PDF documents (e.g., "Daily Absentees" and "Consolidated Monthly Reports").

Batch Data Integration: By utilizing FastAPI's UploadFile class alongside Pandas, the application maps raw column headers from uploaded .csv or .xlsx files, enabling bulk student registrations and batch attendance imports in under a second.

Robust Conflict Resolution: The PostgreSQL backend is instructed to use ON CONFLICT clauses. If a faculty member re-marks attendance for a previously recorded day, the database intelligently overwrites the existing row instead of triggering critical duplicate key errors.

Native File Sharing Integration: The JavaScript frontend integrates the modern Web Share API (`navigator.share`), allowing users to instantly transmit generated PDF reports to external applications (such as Email or WhatsApp) directly from the browser.

7. Conclusion and Future Scope

7.1 Conclusion

The deployment of this FastAPI and PostgreSQL-powered ERP system successfully digitalizes the administrative ecosystem at ISM Patna. By transitioning from isolated, manual spreadsheets to a synchronized, cloud-hosted platform, the institution achieves impeccable data accuracy. The integration of role-specific portals, automated shortage analytics (<75% alerts), and a digital leave approval inbox minimizes administrative bottlenecks and empowers faculty to focus on academic delivery.

7.2 Future Scope

Biometric & RFID Integration: Bridging the system's APIs with physical hardware scanners for frictionless, automated data entry.

Progressive Web App (PWA) Deployment: Utilizing the existing `manifest.json` configurations to deploy the system as a fully standalone, native application for Android and iOS environments.

Predictive Machine Learning Analytics: Implementing time-series forecasting models on the database to identify long-term absenteeism patterns and flag potential student dropouts for early academic intervention.